

Nabd · نبض

Read where the network goes silent.

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Nabd turns the mobile network into a live disaster-impact map. When a disaster strikes, a whole area suddenly goes dark on the network. Nabd reads that silence and tells emergency teams, within the first minute, where the damage is worst and who is cut off.

What we want to build: a tool for disaster command centres that shows, in real time, the map of a disaster's impact — so the first responders go to the worst-hit areas first, instead of waiting hours for the picture to form from scattered phone calls.

THE PROBLEM

In the first hours after a disaster, rescue is held back not by resources but by information: where is the damage worst, and who has been cut off. Today that picture is pieced together slowly — from fragmented emergency calls, social media and field reports — at exactly the moment the network is overloaded and calls cannot get through.

The region carries fresh wounds: the 2023 Kahramanmaraş earthquakes (over 53,000 dead), Morocco's Al Haouz earthquake, Libya's Derna flood, and the 2024 UAE and Oman floods. In each, sensors confirmed *something* happened in seconds — but not which neighbourhoods had lost their people. That map arrived hours late.

WHAT NABD DOES — AND DOES NOT — CLAIM

Nabd is **not** an early-warning system; seismometers already detect the quake. Nabd answers the next question: the **impact map in the first minute, and triage over the first 72 hours**. The insight is simple — when devices across an area fall silent together, the network already knows where the disaster hit. The same signal works for earthquakes, floods, storms and mass power outages.

HOW IT WORKS — TWO LAYERS

1. **Detection.** An AI agent watches area-level network behaviour across neighbouring cells — reachability and congestion, read every 30 seconds. A disaster leaves a signature the network cannot hide: a block of adjacent cells that stops answering at once, while the ring around it saturates as everyone calls at the same time. This uses only anonymous, area-level data — no personal tracking.
2. **Triage.** For people who opted in to be protected — the elderly, the disabled, pilgrimage groups, lone workers — the agent checks who inside the hit area is unreachable and where they were last seen, so rescue teams can find them.
3. **Output.** A live map of the impact and a prioritised list of the worst-hit zones, delivered straight to the command centre and updated as the situation evolves.

TELLING A DISASTER FROM AN ORDINARY OUTAGE

The hard question is not whether an outage can be seen. It is whether a disaster footprint can be told apart from every other reason a group of cells goes quiet. Nabd does not rest on one signature being right. It rests on a **minimum observable pattern** — a contiguous block of cells that stops answering together, in a place where that is not normal — and treats everything else as corroboration that raises confidence rather than as a precondition.

SIGNAL	ROLE	READ FROM
Contiguous silence	Necessary — a connected block, at or above the size floor	Device Reachability
Local baseline	Necessary — silence must be abnormal <i>here</i> , measured against these cells' own history	Device Reachability
Synchronised onset	Corroborating — the block went dark inside one window	Device Reachability
Hot ring	Corroborating — the surviving ring saturates as everyone calls at once	Congestion Insights

Two corroborations make a HIGH-confidence footprint, one makes MEDIUM, none is an abstention with the reason written down. Anything the operator's maintenance calendar already explains is removed before the geometry is even computed. The corroborations are a list, not a hard-coded pair: a new aggregate network surface, when Open Gateway ships one, is added as one more piece of evidence weighed the same way — the detection layer extends rather than being redesigned.

Four look-alikes are demonstrated rather than asserted, each with the agent staying silent and writing down why: a **single-cell fault** (one sentinel down, neighbours normal), a **maintenance window** (the block matches a ticket), a **stadium crowd** (congestion with every sentinel still answering), and the hardest one — **chronic degradation**, a block on a failing backhaul that reproduces every signal an impact has except that it has been happening all morning. That last one is refused by measurement: *these four cells were already unreachable in 3 of the last 11 passes*. In the same run, a real earthquake elsewhere in the city is still declared, which is how the refusal is shown not to be blindness.

WHY THE NETWORK — AND WHY CAMARA

An operator's own control room sees a cell go down, but it sits inside one operator. Emergency agencies, the Red Crescent, municipalities and hospitals are on the outside, and cannot log into three separate operators during a disaster. The CAMARA / Open Gateway network APIs are the one standard, operator-independent way to expose this signal to responders — and to unify every national operator behind a single interface. This is feasible now because these APIs reached stable release in 2025 and are available on the Nokia Network-as-Code platform.

NETWORK APIS USED (NOKIA NETWORK-AS-CODE)

API	ROLE
Congestion Insights	Core — detection: the saturation of the ring around a silent block
Device Reachability Status	Core — detection (sentinels) and triage (who among registered people is unreachable)
Location Retrieval	Core — triage: last-seen position of the unreachable
Geofencing	Roadmap — the footprint is drawn from cell geometry today
Quality on Demand	Roadmap — a resilient responder channel at scale

Three APIs carry the whole workflow, and each one is load-bearing: remove any of them and Nabd stops answering the question it exists to answer. Two more are named as a roadmap rather than built, because a small number of APIs at the centre of the value proposition is worth more than a longer list that is only loosely attached to it.

WHAT IS PROVEN LIVE, AND WHAT IS STAGED

A sandbox cannot stage an earthquake, so Nabd separates the two claims instead of blurring them.

The live platform proves the integration: the same agent runs against real Nokia Network-as-Code devices and every request and response is written to file. **The simulator proves the scenario:** a disaster shaped on the public record of February 2023, run through the identical code path.

The risk in that split is not the simulator — it is a reader suspecting a discontinuity between the two paths. So the continuity is measured rather than asserted, by a harness that a judge can run:

1. **The agent cannot see the backend.** Every module between a response and a decision is parsed and checked to import no SDK, construct no gateway and name no simulated world.
2. **One function chooses the network.** Every runner, scene and harness reaches the platform through a single call; nothing else in the package names a backend.
3. **The live parsers read the simulator's bytes.** Each scene is recorded and replayed through a third backend that reads every response with the same three parsing functions the live gateway uses. The evidence has to come out identical, line for line. If the simulator answered in a shape the live code could not read, this fails.
4. **The live transcript replays with no account.** The recorded platform responses run back through the same agent offline — the check a judge can repeat without credentials.

TESTED AGAINST A REAL EVENT, NOT ONLY AGAINST OUR OWN SCENARIO

One scene is not ours. Its geometry and the intensity of shaking in every cell come from the **USGS ShakeMap for the M7.8 Pazarcik earthquake** of 6 February 2023, 01:17:34 UTC — an intensity field constrained by 262 seismic stations and 1,459 intensity observations. Only the rule that turns shaking into a silent cell is ours, and both of its mechanisms are reported from the event: masts came down with the buildings they were mounted on, and the sites that survived lost mains power and drained their batteries. The thresholds are calibrated against Turkcell's reported *more than half of local base stations inoperative*, and a test keeps them there.

WHAT THE AGENT DID	MEASURED
First map	declared 55 s after onset — 21 contiguous cells, ~2,100 km ² , HIGH confidence
Accuracy	every cell it named is one the measured shaking places above the collapse threshold; no invented damage
What it held back	2 of 23 collapse-band cells — an isolated pocket below the three-cell floor, deliberately not claimed, joining minutes later
How it followed the event	2,100 → 6,300 km ² over 12 updates as batteries failed; 63% of the window dark, against the reported >50%
Triage	22 registered people unreachable at peak, ranked medical-dependent first, every query inside the footprint

This is the difference between a demo that is tuned and one that is checked. The scenario was not shaped to make the agent look good; the shaking is what it was, and the two cells the agent refuses to claim in the first minute are in the record because refusing them is the behaviour we argue for.

WHY A SENTINEL MESH, AND NOT THE PUBLIC'S PHONES

Detection reads a fixed grid of **sentinel devices** — municipal or operator-owned SIMs, one per cell, never a member of the public. A judge is right to ask why that layer needs to exist at all. Three reasons, and the first is not a preference:

1. **The aggregate surface does not exist yet.** The CAMARA endpoints that reached stable release are *device-scoped*: you ask about a device, not about an area. A sentinel per cell is how an area-level reading is synthesised from the APIs that actually ship today. If Open Gateway later publishes a true area API, it enters Nabd as one more signal in the same evidence list — which is why that list is built to be extended.
2. **It is what makes the anonymous layer genuinely anonymous.** The alternative — sampling real subscribers to infer that an area has gone dark — would query people who never opted in. With sentinels, the detection layer touches no member of the public at all, ever, and that is a property of the device list rather than a promise about behaviour.
3. **A fixed device is a better sensor than a phone.** A handset that stops answering may have moved, run flat or gone into a lift. A traffic cabinet, a water-pump telemetry unit or a streetlight controller did not walk away: when it goes quiet, something happened to the cell or to the site.

This is also less deployment than it sounds. Municipalities already operate large fleets of SIM-equipped fixed assets. Nabd needs a list of them and permission to query them — not a construction project.

THE AT-RISK REGISTRY — HOW IT IS HELD (STATED ASSUMPTION)

The triage layer needs a list of people who chose to be findable. Three models are possible: the subscriber consents through their operator; the **disaster agency holds the register and operators honour it**; or an emergency legal basis removes per-person consent altogether. Nabd assumes the second, and this is an assumption we have adopted deliberately, not a settled fact.

TEST	WHY THE AGENCY-HELD REGISTER WINS IT
Fewest parties	One agency and the operators it already coordinates with in an emergency — no new consent flow inside every operator's product
Clearest accountability	The register is a duty-of-care list, held by the body that already owes that duty and is already audited on it
Easiest to explain	A citizen opts in once, with the agency they would call anyway; the operator's role is to honour a list, not to interpret one
Survives national scale	The register sits above the operators, so it still works when the people in one footprint are spread across three networks

We have deliberately *not* built the demonstration around emergency legal powers. That path may well be the one a country chooses, but it moves the conversation from what the product does to a policy debate that cannot be settled inside a hackathon.

USERS & PRIVACY

Built for national disaster agencies and search-and-rescue organisations — not a consumer app. Detection reads only anonymous, area-level data; the triage layer touches an individual only when that person has opted in *and* their home cell sits inside a footprint the agent has already declared. No cameras, no message content, no tracking of the public.

That boundary is enforced by the shape of the agent rather than by a setting. The node that queries a personal device is reachable only from a verdict with an active footprint, so the claim is a property of the topology; a fuzz test drives thousands of random passes and checks that every personal-device call belongs to a registered person inside the footprint; and every pass of the evidence file states, in the record itself, whether the personal-data path was open and how many personal calls it made. On an ordinary morning that number is zero on every line.